

NHL Beyond Age 27: When Do Players Peak? — A Corsi-Based Analysis

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Why this work.

Under a hard salary cap, teams routinely commit multi-year dollars to skaters entering their late 20s. Small errors in estimating **peak age** and post-peak decline can lock in cap inefficiencies. We focus on skaters (F/D; goalies excluded) at **even strength**, using **Corsi** as an on-ice performance proxy, to estimate *when* players peak and *how quickly* performance changes beyond that point..

What prior studies show.

Our study is motivated by Elijah Cavan, Jiguo Cao, and Tim B. Swartz's paper, *NHL Aging Curves using Functional Principal Component Analysis* (2025), which analyzes NHL players from 1920–2022 (restricted to ages 22–34 and careers ≥ 7 seasons) and finds **concave aging curves peaking around 26–28**, i.e., roughly age 27. Related hockey analytics work—e.g., **Evolving-Hockey** aging-curve analyses, **Eric Tulsky's** early shot-rate/aging posts, and **Rob Vollman's *Hockey Abstract*** chapters on age and contract value—likewise place peak performance in the mid-to-late 20s, with role-dependent decline.

What we will do (our contribution).

We build **five-year player panels** aligned on each skater's **peak season** (relative ages $-2 \dots +2$), compute **unweighted** and **role-weighted composite z-scores** from shot-attempt metrics, and compare **distributions**, **trend lines with 95% CIs**, and **regressions** controlling for role and relative age. The goal is to provide **evidence-based guidance** for roster and cap decisions **beyond age 27**.

What is Corsi (our performance proxy).

Corsi counts all shot attempts at even strength—shots on goal, missed shots, and blocked shots. We use **Corsi For (CF)** and **Corsi Against (CA)** to summarize whether a team takes more attempts than it allows while a player is on the ice. The common rate is $CF\% = CF / (CF + CA)$. Because shot attempts are frequent and predictive at the team level, Corsi is a standard proxy for **driving play** (possession/territory). We analyze skaters at **even strength** (goalies excluded).

Data Sources



Evolving-Hockey (Primary Data Source)

- **What it is:** Curated historical NHL skater and team performance metrics.
- **How we used it:** Exported to CSV and loaded into our Postgres database (existing tables).
- **Key fields used:** `player`, `eh_id`, `api_id`, `season`, `team`, `position`, `shoots`, `birthday`, `age`, `draft_year`, `draft_rnd`, `draft_overall`, `games_played`, `time_on_ice`, `GF%`, `SF%`, `FF%`, `CF%`, `xGF%`, `GF/60`, `GA/60`, `SF/60`, `SA/60`, `FF/60`, `FA/60`, `CF/60`, `CA/60`, `xGF/60`, `xGA/60`, `G+/-/60`, `S+/-/60`, `F+/-/60`, `C+/-/60`, `xG+/-/60`, `Sh%`, `Sv%`.
- **Coverage period:** 2013–14 through 2024–25 (season key uses the end year; e.g., **2014** = 2013–14).
- **Volume:** 3,126 skater–season rows after cleaning and exclusions.

Hockey-Reference (Secondary) –

<https://www.hockey-reference.com/>

- **What we scraped.** Player regular-season pages for even strength and all situations. Goalies.
- **Formats.** HTML tables → **CSV**.
- **Key variables actually used.** `player_name`, `player_href` (stable URL id), `birth_year` (when available), `position`, `season`, `team(s)`, `TOI (EV/total as available)`. These support **ID normalization**, traded-season **totals**, and **eligibility filters** (min TOI).
- **How many records.** 29,000 skaters; 1.2k goalies.
- **Time period covered.** Seasons 2013–14 - 2024–25, inclusive.

Spotrac NHL (Tertiary) – <https://www.spotrac.com/nhl/contracts>

- **What we scraped:** player cap/contract data, CSV →
- Seasons 2013–14 through 2024–25 (inclusive).

Data Manipulation

EVOLVING HOCKEY prep (SQLAlchemy)

- **Typed load:** Declared tables with explicit types (`String`, `Integer`, `Numeric(5,2)`) and loaded cleaned CSVs via `engine.begin()` + `insert()`.
- **Season & totals:** Standardized season to **end-year**; collapsed multi-team seasons into one row using **TOI-weighted** sums (implemented in a `CREATE VIEW` executed via SQLAlchemy).
- **Eligibility & QC:** `WHERE` filters (non-null CF/CA/TOI, min TOI/games, positions F/D only).

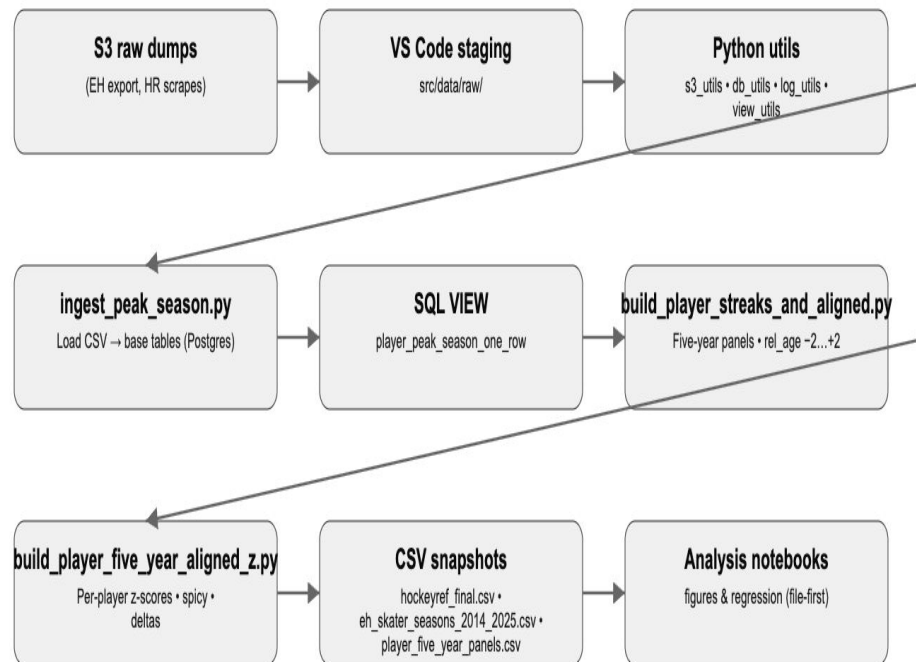
Panel assembly (SQLAlchemy)

- **Five-year window:** Built peak-aligned panels (`rel_age = -2...+2`) with window functions; kept only players with a full 5-year streak.

Z-scores & composites (SQLAlchemy)

- **Per-player z-scores:** Computed `cf_pct_z`, `cf60_z`, `ca60_z` using window `AVG/STDDEV_SAMP` and `CASE` to null out zero-std cases.
- **“Spicy” scores:** Produced **unweighted** and **role-weighted** composites with `CASE` (different weights for D vs F).
- **Peak deltas:** Compared each `rel_age` to `rel_age=0` via `MAX(CASE WHEN rel_age=0 THEN ... ) OVER (...)` and subtraction.

NHL-Beyond-27 — Data Architecture



Data Manipulation II

Scrape & stage: Pulled **regular-season skater pages**, **even-strength tables**, and **goalie pages**; saved season CSVs into `data/seasons/` and `data/even_strength/`.

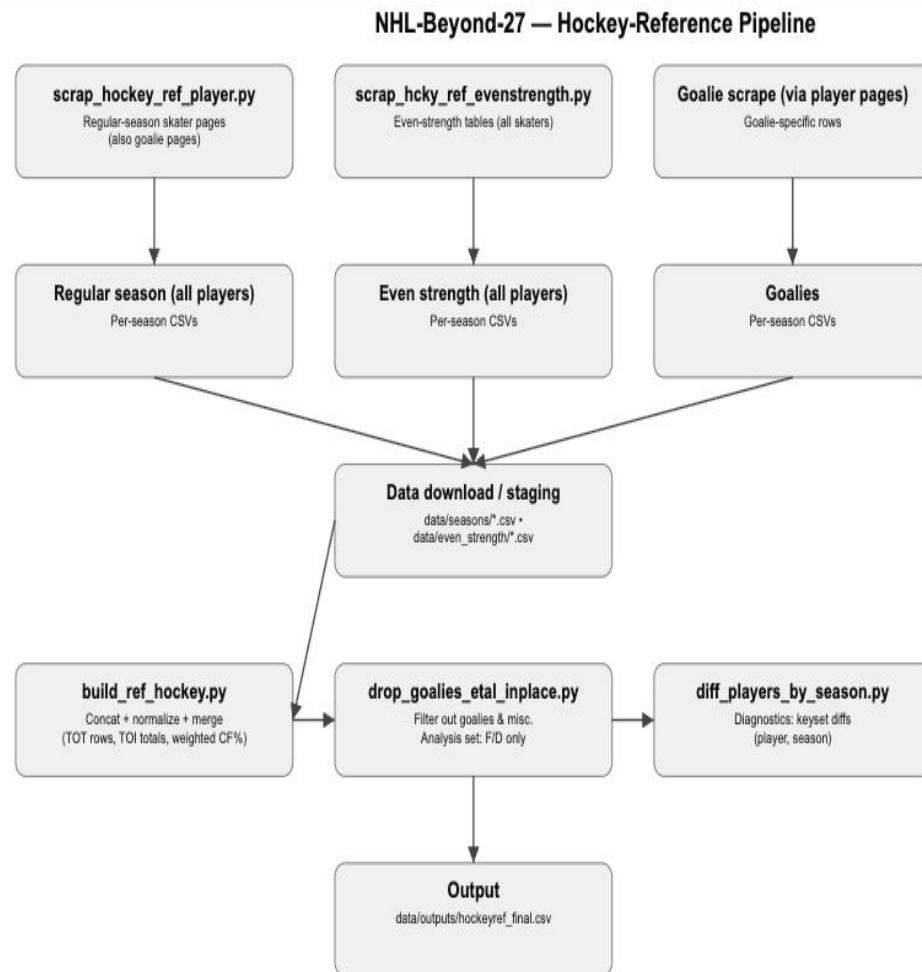
Normalize keys: Standardized columns (**player**, **season**, **tm**, **pos**), lowercase names / uppercased team codes, unified season strings to end-year.

Handle multi-team seasons: If an **nTM/TOT** row exists, keep it; otherwise **aggregate team rows** to a synthetic **TOT** by **summing numerics** and taking **mode(pos)**.

Time-on-ice math: Converted per-game TOI (**mm:ss**) $\times$ GP $\rightarrow$ **season totals** (**toi_seconds_total**, **toi_total_hms**); left even-strength **uncollapsed** by team except for merge-time totals.

Quality filters & dedupe: Dropped goalies for analysis set, enforced **min TOI/games**, removed duplicates on (**player**, **season**), and guarded rate calcs with **NULLIF** to avoid divide-by-zero.

Merge & export: Built a temporary (**player**, **season**) even-strength aggregate (TOI-weighted CF% where available), **merged** with standard table, and wrote `data/outputs/hockeyref_final.csv` for downstream use.



Data Manipulation III

Why Hockey-Reference was hard to scrape

Sports-Reference deliberately makes automated collection difficult.

What we did after scraping:

Normalize & clean:

- Standardized names (case/whitespace), positions, and team strings.
- Converted times (e.g., TOI) and dates to canonical numeric formats.
- Harmonized season keys (end-year convention: 2014 = 2013–14).

Make the rosters line up

- Built a **player mapping** to reconcile duplicates, aliases, and diacritics (e.g., “Mikko Rantanen” vs “M. Rantanen”).
- Dropped partial seasons and clearly mismatched rows to ensure a like-for-like population.

Use a **diff** script to resolve mismatches

- Ran a simple **diff** over player lists from Hockey-Reference and our Evolving-Hockey export.
- Flagged **only** the disagreements (missing, extra, or renamed players), then iterated until the sets matched.

Hockey-Reference vs Evolving-Hockey — Player Set Overlap

Metric	Count
Hockey Reference Total	288
Evolving-Hockey Total	282
Intersection(normalized matches)	264
HR Only(In HR not in EH)	24
EH Only(In EH not in HR)	18
Union(unique players across both)	306

About 1 in 3 players show a clear performance peak around age 27



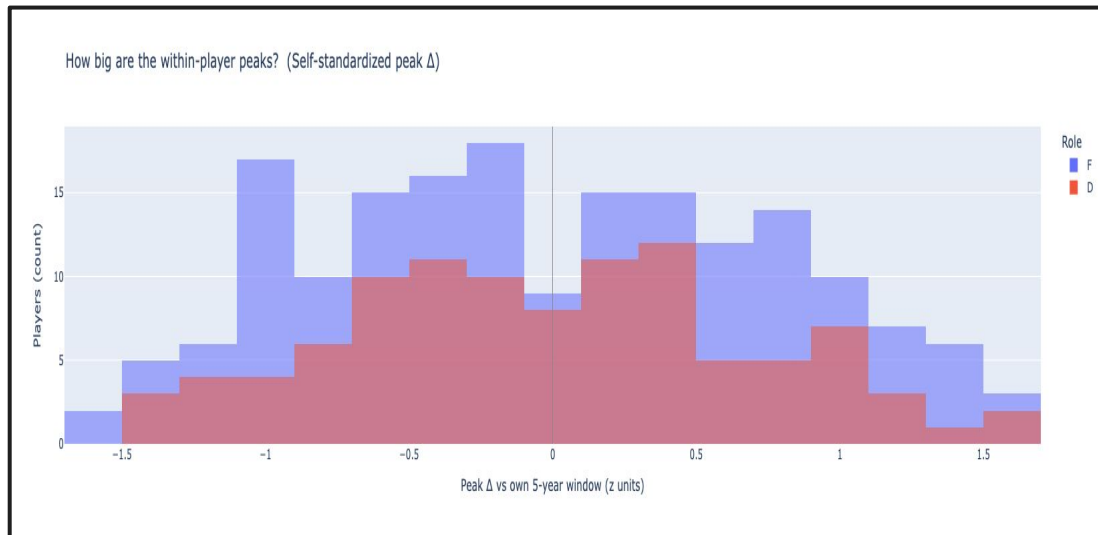
Timing: Peaks are **centered near 27** ($\text{rel_age} \approx 0$), not uniformly spread.

Role split: Forwards peak a bit more often than Defense (37% vs 32%).

Context: A peak relative to one's own history doesn't imply separation from peers that year (see Quadrant page: $r \approx 0$).

Metric*	Role	n	n_peak	share_peak_%
cf60_dz	D	102	33	32.4
cf60_dz	F	180	67	37.2

*cf60_dz: delta change by rel. age corsi for minute normed



Based on CF60 z-score changes across a 5-season window ($\text{rel_age} \in \{-2, -1, 0, 1, 2\}$), 35.5% of players (100/282) exhibit a **concave peak** with vertex in $\{-1, 0, 1\}$; Forwards: 37.2%, Defense: 32.4%.

Concave peak test: fit $y = ax^2 + bx + c$ on rel_age points; flag **peak** when $a < 0$ and vertex $x^* = -b/(2a) \in \{-1, 0, 1\}$ ($\pm$ small tolerance).

Does Weighting the Score Tilt the Ice?

We experimented with the Corsi Formula by introducing a heuristic weighting component to the scores of defensemen and offensive players. Is it possible to correct any negative bias given to forwards by the corsi against and to defensemen by corsi for?

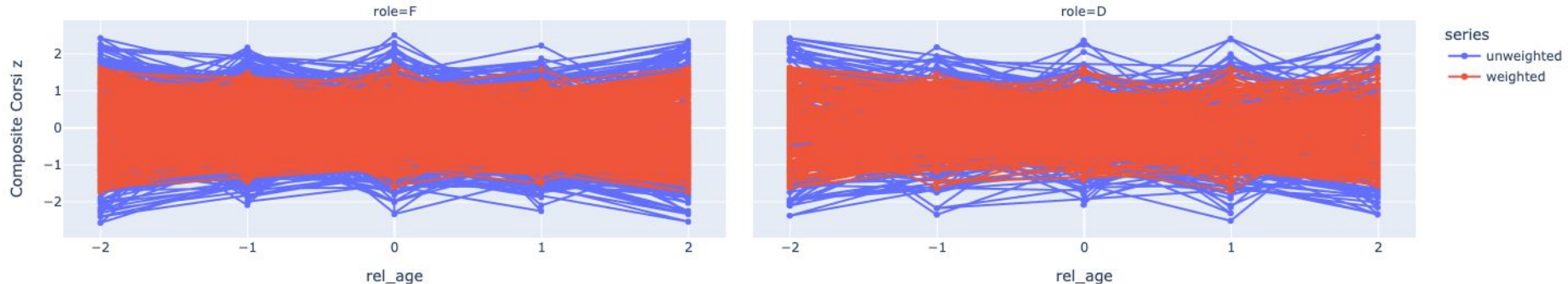
--- Role-weighted composite z (heuristics)

D: $0.5*cf_pct_z + 0.2*cf60_z - 0.3*ca60_z$

F: $0.5*cf_pct_z + 0.3*cf60_z - 0.2*ca60_z$

As you can see in the chart below, there is no significant change in the data. The variance was reduced without shifting the center, mean value. Weighting acts like a mild regularizer but does not change the story. This is further supported by pearson_r score of .996 and a mean oy ~ 0.000.

Composite Corsi z: weighted vs unweighted (by rel_age, faceted by role)



Peak Δ — Top-5 “pop above your own norm”

For any metric:

Peak Δ = value at `rel_age` = 0 (the peak season) minus the player’s own 5-year average over `rel_age` = -2...+2.

It answers: “How much did this player rise above their personal baseline at their peak?”*

What’s plotted here:

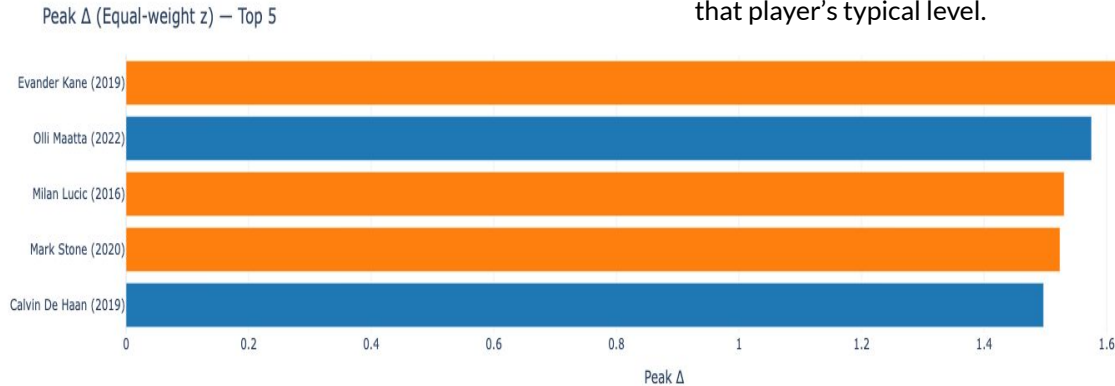
Bottom chart: Top 5 players by Peak Δ (largest positive lift vs. their own baseline).

In this panel we’re using **CF60_Z** (shot creation per 60, standardized within player).

Longer bars = bigger one-season jump above that player’s typical level.

Each player’s cap hit (USD) across the same 5-year window

player	peak_year	rel_age	cap_hit_usd
Milan Lucic	2015	-2	\$6,000,000.00
Milan Lucic	2015	-1	\$6,000,000.00
Milan Lucic	2015	0	\$6,000,000.00
Milan Lucic	2015	1	\$6,000,000.00
Milan Lucic	2015	2	\$6,000,000.00
Evander Kane	2018	-2	\$5,250,000.00
Evander Kane	2018	-1	\$7,000,000.00
Evander Kane	2018	0	\$7,000,000.00
Evander Kane	2018	1	\$7,000,000.00
Evander Kane	2018	2	\$2,108,696.00
Calvin De Haan	2018	-2	\$3,300,000.00
Calvin De Haan	2018	-1	\$4,550,000.00
Calvin De Haan	2018	0	\$4,550,000.00
Calvin De Haan	2018	1	\$4,550,000.00
Calvin De Haan	2018	2	\$4,550,000.00
Mark Stone	2019	-2	\$7,350,000.00
Mark Stone	2019	-1	\$9,500,000.00
Mark Stone	2019	0	\$9,500,000.00
Mark Stone	2019	1	\$9,500,000.00
Mark Stone	2019	2	\$9,500,000.00



Reading tip: A big positive bar means the player didn’t just have a good year—they outperformed their own norm in a meaningful way.

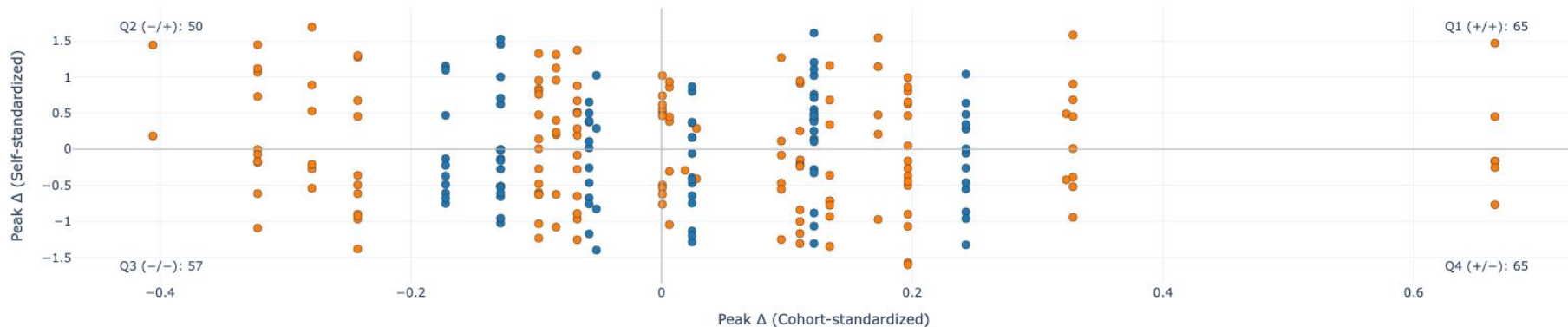
Self vs Cohort Peak Performance (Quadrant View)



Quadrant counts:

- Q1 (+/+): above-average vs. peers *and* above their own window.
- Q2 (-/+): below peers, but above their own window.
- Q3 (-/-): below peers and below own window.
- Q4 (+/-): above peers, but below own window.

Self vs Cohort Peak Δ — Weighted ($r = -0.03$ [-0.16,0.09], $n=237$)



“Self vs Cohort Peak Δ are largely orthogonal ($r \approx 0$). Players who **spike vs their own history** aren’t systematically the same ones who **separate from peers** that year. Consistent elites stay elite (small self Δ , big cohort Δ), while volatile players can pop relative to themselves without clearing peer bars. Role/usage, context, and noise further decouple the two.”

Peaks are personal — CF60 z peaks near age ~27

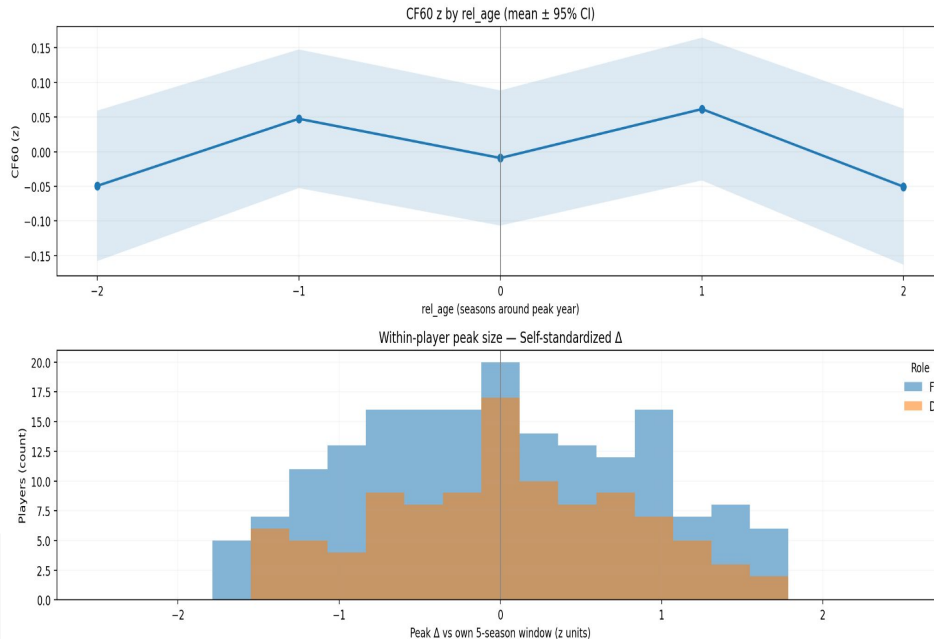
Across a 5-season window ($\text{rel_age} \in \{-2, -1, 0, 1, 2\}$), players' CF60 z-scores are significantly higher at $\text{rel_age} = 0$ than surrounding seasons, indicating a statistical peak in athletic output (shot-creation pace).

- **Evidence:** $\text{rel_age} = 0$ outperforms adjacent years ($-1, +1$) on CF60 z (cluster-robust $p < 0.05$, by player).
- **Scope:** holds for the full sample and within roles (F/D), with small role differences in magnitude.
- **Context:** “Peak vs self” doesn’t imply “separation from peers” in the same year (Self Δ vs Cohort Δ : $r \approx 0$).

Chart Notes

*Window = five seasons centered on peak year;

**CF60 z standardized within role $\times$ season. Inference uses player-clustered SEs.



Contributions



- **Contributions and Authorship.** The study was initially conceived jointly by Eric Winiecke and Chris Halvonik, including the high-level research questions and proposed data sources. During implementation, the collaborators adopted different technical stacks (Eric: Visual Studio Code, GitHub, and PgAdmin with a local Postgres database; Chris: Google Colab), which introduced architectural incompatibilities and hindered coordinated development. Additional complexity associated with extracting and standardizing Hockey-Reference data further widened the methodological paths. In light of these constraints, Eric proceeded with the original proposal and completed an independent implementation and analysis; this submission reflects Eric Winiecke's sole authorship of the final code, data processing, and results.